

# Measurement of $\nu_\mu$ Charged-Current Multi-Pion Production on Argon with the MicroBooNE Detector in the NuMI Beam

Technical Note — Two- and Three-Charged-Pion Final States (CC1mu2pi, CC1mu3pi)

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## Abstract

We describe the extension of the MicroBooNE NuMI  $\nu_\mu$  charged-current single-charged-pion analysis to higher pion multiplicities: charged-current interactions producing a muon and exactly two or three charged pions (and any number of protons) in the final state. These  $N\pi^\pm$  topologies probe the resonant and deep-inelastic regimes and the intranuclear pion cascade (final-state interactions), and are a significant background to the single-pion signal. The selections are built as thin subselections of the inclusive single-pion selection, adopting the looser pion identification of the dedicated MicroBooNE multi-pion study and replacing its log-likelihood pion cut with a dedicated boosted decision tree (BDT). On the Run-1 forward-horn-current  $\nu_\mu$  overlay the two-pion (three-pion) selection reaches a signal efficiency of 17.9% (10.1%) at an intrinsic purity of 26.8% (11.3%), corresponding to of order 670 (66) selected signal events at full exposure. We present the selection, the pion-identification BDT, the cut-flow, and a study of the pion-momentum thresholds that set the accessible signal phase space; the two-pion channel supports a flux-averaged total cross section and a coarse differential measurement, while the three-pion channel is a total-cross-section measurement. This note is a companion to the single-pion analysis note, from which the detector model, samples, exposure, systematic framework and unfolding are inherited unchanged.

# Contents

|    |                                                         |   |
|----|---------------------------------------------------------|---|
| 1  | Introduction                                            | 2 |
| 2  | Signal definition                                       | 2 |
| 3  | Samples, exposure and normalisation                     | 2 |
| 4  | Event selection                                         | 3 |
| 5  | Selection performance                                   | 3 |
| 6  | Pion-momentum thresholds and the accessible phase space | 4 |
| 7  | Backgrounds                                             | 6 |
| 8  | Feasibility and observables                             | 6 |
| 9  | Systematic uncertainties                                | 6 |
| 10 | Status and follow-up                                    | 6 |

# 1 Introduction

Charged-current (CC) production of multiple charged pions on nuclei sits at the transition between the resonant (RES) and deep-inelastic (DIS) regimes of neutrino–nucleus scattering and is strongly shaped by the intranuclear hadronic cascade. A  $\nu_\mu$  interaction that produces several pions at the primary vertex may have some of them absorbed, charge-exchanged or produced in re-scattering before they leave the nucleus, so the observed charged-pion multiplicity is a direct probe of final-state interactions (FSI). Multi-pion final states are also a non-negligible background to exclusive single-pion and pion-less measurements, so a dedicated measurement of their rate and kinematics on argon is valuable both as a cross-section result and as a constraint on the interaction model used across the MicroBooNE programme.

This note documents the extension of the NuMI  $\nu_\mu$  CC  $1\pi^\pm$  analysis (the “single-pion note”, to which this is a companion) to the two- and three-charged-pion final states. The selections, `CC1mu2pi` and `CC1mu3pi`, are implemented as subselections of the inclusive single-pion selection `CC1mu1piXp` in the `xsec_analyzer` framework and reuse its muon reconstruction, samples, per-run POT normalisation, systematic universes and Wiener-SVD unfolding without modification. The multi-pion pion identification follows the dedicated MicroBooNE multi-pion study of Mahmud [1] and the accompanying analyzer of Gu [2].

Given the steeply falling multi-pion cross section and the reduced reconstruction efficiency for events with several (often soft) pions, these channels are statistically limited: the emphasis of this first note is the selection definition, its efficiency and purity, the accessible phase space, and a feasibility assessment for total and differential cross sections.

## 2 Signal definition

The signal is a  $\nu_\mu$  (or  $\bar{\nu}_\mu$ ) charged-current interaction with a true muon and *exactly*  $N$  charged pions ( $\pi^+$  or  $\pi^-$ ; MicroBooNE does not determine the pion charge sign) in the final state, with  $N = 2$  (`CC1mu2pi`) or  $N = 3$  (`CC1mu3pi`). The signal is inclusive with respect to protons, neutrons and neutral pions/electromagnetic activity, subject to the requirement that there is exactly one muon and no other charged leptons. The interaction vertex must lie in the fiducial volume defined in the single-pion note.

**Pion-momentum threshold.** The accessible signal phase space is bounded from below by the momenta of the pions themselves: a pion must be energetic enough to produce a reconstructable track. In a genuine two- or three-pion event the sub-leading pions are systematically softer than the leading one, and the softest pion sets the floor on how often the full topology can be reconstructed. The single-pion signal definition applies a threshold to *the* pion; for the multi-pion channels this must be generalised so that the momentum requirement applies consistently to *all*  $N$  pions rather than to one. Section 6 quantifies the true leading and sub-leading pion spectra and studies the selection efficiency as a function of a per-pion momentum threshold, and recommends the threshold that defines the measurable phase space. That study (Section 6) adopts a per-pion threshold of  $p_\pi > 0.10 \text{ GeV}/c$  applied to *all*  $N$  pions (the pion tracking turn-on), together with the muon threshold  $p_\mu > 0.15 \text{ GeV}/c$  inherited from the single-pion definition.

## 3 Samples, exposure and normalisation

The samples, beam exposure, per-run protons-on-target normalisation, beam-off (EXT) cosmic treatment and dirt overlay are identical to the single-pion note and are not repeated here. All results in this draft use the Run-1 forward-horn-current (FHC)  $\nu_\mu$  overlay for the efficiency and cut-flow studies; the extension to the full FHC, RHC and combined exposures follows the same

per-run machinery. As in the parent analysis, the beam-on data remain blinded and all data-like distributions use Poisson-thrown fake data.

## 4 Event selection

`CC1mu2pi` and `CC1mu3pi` inherit the complete `CC1mu1piXp` selection — the same slice, fiducial-volume, topology and muon-candidate requirements — and the same reconstructed observables. Only the pion identification and the required charged-pion multiplicity differ. In the code the multiplicity is a virtual (`required_charged_pions()`) so that the derived value ( $N = 2, 3$ ) takes effect inside the inherited signal definition and reco selection.

Re-using the single-pion pion identification directly gives a prohibitively low efficiency (4.2% for  $2\pi$ , 0.8% for  $3\pi$ ), because that identification was tuned to isolate a single energetic pion and reject protons: a tight 4 cm vertex window, a track-multivariate/Bragg/length requirement, and cuts on the single muon–pion opening angle and on electromagnetic showers. Following Mahmud [1] and Gu [2], the multi-pion selections loosen the pion identification through five parameters, implemented as overridable virtuals on the base class so that the single-pion selection is left byte-for-byte unchanged (Table 1):

Table 1: Pion-identification parameters overridden by the multi-pion selections. The inclusive defaults reproduce the single-pion selection exactly.

| Parameter                       | Single-pion default             | Multi-pion   |
|---------------------------------|---------------------------------|--------------|
| Vertex-to-track window          | 4 cm                            | 9.5 cm       |
| Pion identification             | LLR-PID + TMVA + Bragg + length | LLR-PID only |
| Uncontained pions allowed       | 0                               | 1            |
| $\mu$ – $\pi$ opening-angle cut | applied                         | not applied  |
| Shower ( $\pi^0$ ) veto         | applied                         | not applied  |

The wider vertex window recovers pions from secondary scatters and slightly displaced starts; the log-likelihood-ratio-only identification drops the proton-rejection cuts that were specific to the one-pion topology; the “maximum uncontainment” tolerance allows a single pion per event to leave the detector; and the opening-angle and shower cuts, which are one-pion-topology cuts (a single  $\mu$ – $\pi$  pair, and an electromagnetic veto), are dropped because they are ill-defined or unwanted for a multi-pion, EM-inclusive signal. In the pion loop the contained and uncontained pion candidates are tallied separately and the charged-pion count is formed as the number of contained pions plus up to one uncontained pion.

## 5 Selection performance

Table 2 gives the cut-flow for both selections on the Run-1 FHC  $\nu_\mu$  overlay. The dominant efficiency loss is at the  $N$ -pion identification stage (`MuonCandidate`→`ContainedPions`): reconstructing  $N$  pion-like tracks is the efficiency ceiling, while the upstream vertex/topology/muon requirements and the downstream wire-gap and track-multiplicity cuts are comparatively gentle. The loosened identification raises the efficiency by a factor of  $\sim 4$  ( $2\pi$ ) and  $\sim 8$  ( $3\pi$ ) relative to the single-pion identification, turning  $3\pi$  from a non-measurement ( $\mathcal{O}(1)$  event) into a viable one ( $\sim 40$  events).

Table 2: Multi-pion cut-flow yields (Run-1 FHC  $\nu_\mu$  overlay, POT-weighted; beam-off cosmics and dirt not included). “Total” is signal + MC background; Eff. is the cumulative signal efficiency relative to the generated signal (None stage); Pur. = Signal/Total and is MC-intrinsic (an upper bound — the real-sample purity is lower once EXT cosmics, which dominate at these multiplicities, are included). The shower veto and opening-angle cut are not applied.

| Cut                                       | Total        | Signal     | Bkg MC       | Eff. [%]    | Pur. [%]    |
|-------------------------------------------|--------------|------------|--------------|-------------|-------------|
| <i>CC1mu2pi — exactly 2 charged pions</i> |              |            |              |             |             |
| None                                      | 759 496      | 3 746      | 755 750      | 100.0       | 0.5         |
| InFiducialVol                             | 177 783      | 3 145      | 174 639      | 83.9        | 1.8         |
| Topological                               | 83 662       | 2 565      | 81 097       | 68.5        | 3.1         |
| MuonCandidate                             | 58 791       | 2 273      | 56 518       | 60.7        | 3.9         |
| ContainedPions                            | 3 372        | 781        | 2 591        | 20.9        | 23.2        |
| MuonIn3Planes                             | 3 181        | 737        | 2 444        | 19.7        | 23.2        |
| PionIn3Planes                             | 2 786        | 652        | 2 134        | 17.4        | 23.4        |
| <b>Final</b>                              | <b>2 500</b> | <b>579</b> | <b>1 921</b> | <b>15.5</b> | <b>23.2</b> |
| <i>CC1mu3pi — exactly 3 charged pions</i> |              |            |              |             |             |
| None                                      | 759 496      | 656        | 758 840      | 100.0       | 0.09        |
| InFiducialVol                             | 177 783      | 558        | 177 225      | 85.0        | 0.3         |
| Topological                               | 83 662       | 457        | 83 205       | 69.6        | 0.5         |
| MuonCandidate                             | 58 791       | 393        | 58 398       | 59.9        | 0.7         |
| ContainedPions                            | 447          | 53         | 394          | 8.1         | 11.8        |
| MuonIn3Planes                             | 415          | 50         | 365          | 7.6         | 12.0        |
| PionIn3Planes                             | 384          | 47         | 337          | 7.2         | 12.2        |
| <b>Final</b>                              | <b>344</b>   | <b>40</b>  | <b>304</b>   | <b>6.1</b>  | <b>11.6</b> |

These efficiencies sit below the values reported in the dedicated study [1] (37% for  $2\pi$ , 31% for  $3\pi$ ) because the selections here retain the inherited muon-momentum-quality, wire-gap, track-multiplicity and software-trigger requirements that the leaner study omits; the trade is a lower efficiency for a markedly higher intrinsic purity, which is the more favourable operating point for a cross-section extraction.

## 6 Pion identification: from LLR to a dedicated BDT

The efficiency numbers above are limited by the per-track pion identification, not by reconstruction. Counting how often each true final-state charged pion is reconstructed as a track at all – independent of whether it survives the pion ID – sets a reconstruction ceiling of 66.6% ( $2\pi$ ) and 52.4% ( $3\pi$ ): every one of those pions is on the table if the particle ID does not discard it. The gap between that ceiling and the realised efficiency is therefore a pure particle-ID loss, which motivates replacing the inherited loose log-likelihood-ratio (LLR) cut with a dedicated classifier.

A boosted decision tree (BDT, TMVA gradient boost) was trained to separate true charged pions from the tracks that pollute the multi-pion sample. The input features are the ones that carry pion-vs-proton/muon information without encoding momentum directly: the pion and proton three-plane LLR scores, the Bragg-peak proton, muon and MIP likelihoods, the track score, length and vertex distance. On a held-out sample the classifier reaches an integrated ROC of 0.81. Truth-matching the counted pions (via the backtracked PDG) shows the residual background is dominated by protons mis-reconstructed as pions ( $\sim 31\%$ ), with real out-of-topology pions ( $\sim 44\%$ ) and muons ( $\sim 12\%$ ) making up the rest; the proton contamination is the irreducible core, since a proton that ranges out exactly like a pion is genuinely ambiguous. Consistent with this, mixing an explicit Bragg-peak proton veto back on top of the BDT did not reduce the proton fraction (the ambiguous protons pass the Bragg cut too) and only cost signal efficiency, so the final configuration uses the BDT alone.

Because the two selections have different soft-pion content, the classifier is deployed differently in each. The  $2\pi$  selection uses two momentum-specialised BDTs, split at a reconstructed track length of 20 cm, with a tighter operating point on the soft branch to protect the exact-multiplicity purity; the  $3\pi$  selection keeps a single BDT, because  $3\pi$  events genuinely contain soft pions and the tight soft cut would erode an already scarce efficiency. Relative to the loose-LLR baseline, the BDT raises the operating point to a signal efficiency of 17.9% at 26.8% purity for  $2\pi$  (split BDT) and 10.1% at 11.3% purity for  $3\pi$  (single BDT), the efficiency for  $3\pi$  improving by roughly two-thirds over the LLR-only selection. The same classifier was trialled on the inclusive  $CC1\mu\pi^\pm Xp$  selection and *hurt* it – efficiency rose from 16.7% to 18.9% but purity collapsed from 60.3% to 38.8% – so the inclusive selection retains its LLR cut and the BDT is used only in the multi-pion subselections.

## 7 Pion-momentum thresholds and the accessible phase space

The efficiency ceiling at the  $N$ -pion identification stage is driven by the softest pions in the event. The signal multiplicity is currently defined by counting charged pions above a negligible momentum ( $p_\pi > 0$ ), so a true  $N$ -pion event is counted as signal however soft its sub-leading pions are — including pions too soft to produce a reconstructable track. Such events enter the efficiency denominator but can never enter the numerator, mechanically depressing the quoted efficiency. A physically meaningful signal definition applies a momentum threshold to *all*  $N$  pions, restricting the measurement to a phase space that can actually be reconstructed.

**Method.** For each true  $N$ -pion signal event we record the leading (hardest) and minimum (softest) true charged-pion momenta, `mc_pionpm_lead_mom` and `mc_pionpm_min_mom`. The efficiency as a function of a per-pion threshold  $p_\pi^{\min}$  is then evaluated on the sub-sample whose *softest* pion exceeds the threshold,

$$\varepsilon(p_\pi^{\min}) = \frac{N(\text{selected} \wedge \text{signal} \wedge p_\pi^{\text{soft}} > p_\pi^{\min})}{N(\text{signal} \wedge p_\pi^{\text{soft}} > p_\pi^{\min})},$$

which isolates the effect of removing unreconstructable soft-pion events from the signal definition.

**Results.** The sub-leading pions are indeed substantially softer than the leading one: on the Run-1 FHC overlay the mean leading (softest) true charged-pion momentum is 0.52 (0.27) GeV/ $c$  for  $2\pi$  and 0.57 (0.21) GeV/ $c$  for  $3\pi$  (Table 3). A significant fraction of events have a pion below the tracking turn-on: 10% (18%) of  $2\pi$  ( $3\pi$ ) signal events have their softest pion below 0.10 GeV/ $c$ , rising to 30% (51%) below 0.175 GeV/ $c$ . These events, with an unreconstructable pion, are counted as signal at the current  $p_\pi > 0$  threshold but can never pass the reco selection.

Table 3: Signal efficiency as a function of a per-pion momentum threshold  $p_\pi^{\min}$  (applied to all  $N$  pions via the softest one), Run-1 FHC  $\nu_\mu$  overlay. “Kept” is the fraction of the  $p_\pi > 0$  signal retained; Eff. is the efficiency evaluated within the restricted phase space.

| $p_\pi^{\min}$ [GeV/ $c$ ] | CC1mu2pi |          | CC1mu3pi |          |
|----------------------------|----------|----------|----------|----------|
|                            | Kept     | Eff. [%] | Kept     | Eff. [%] |
| 0 (pre-fix)                | 1.00     | 16.3     | 1.00     | 6.4      |
| 0.10                       | 0.90     | 17.1     | 0.82     | 6.2      |
| 0.175                      | 0.70     | 18.9     | 0.49     | 8.1      |
| 0.25                       | 0.43     | 20.2     | 0.26     | 5.6      |

The central result is that raising the threshold *barely changes the efficiency*: restricting to events whose softest pion exceeds 0.175 GeV/ $c$  — i.e. where all pions are comfortably above

tracking turn-on — still yields an efficiency of only  $\sim 19\%$  ( $2\pi$ ). Roughly four in five events with fully reconstructable pions are still lost. The efficiency ceiling is therefore *intrinsic multi-track reconstruction* — resolving  $N$  distinct pion tracks from a common vertex in the presence of hadronic re-interactions and track merging — and not the soft-pion threshold. The soft sub-leading pions are a real but secondary effect (10–18% of events genuinely below tracking threshold).

**Recommendation.** A per-pion momentum threshold of  $p_{\pi}^{\min} \simeq 0.10 \text{ GeV}/c$  (the pion tracking turn-on) applied to all  $N$  pions is adopted for the signal definition: it removes the genuinely unreconstructable soft tail (10–18% of events) at negligible efficiency cost, defines an honest measurable phase space, and preserves  $\sim 90\%$  ( $2\pi$ ) /  $\sim 82\%$  ( $3\pi$ ) of the signal — important given the limited statistics, especially for  $3\pi$ . A tighter  $0.175 \text{ GeV}/c$  threshold, matching the single-pion measured phase space, would be cleaner but sacrifices half of the  $3\pi$  signal and is not justified by the marginal efficiency gain. The threshold does not, and is not expected to, recover the bulk of the efficiency, which is limited by reconstruction and is corrected for by the unfolding.

The threshold scan above was performed on the pre-fix signal definition, which also contained a bookkeeping error: the momentum and opening-angle requirements were applied to `TrueCandidatePionP`, the *last* charged pion encountered in the daughter loop, rather than to a well-defined pion. For the single-pion signal this is unambiguous (there is only one pion), but for  $N > 1$  it applied the cuts to an arbitrary, ordering-dependent pion. Both issues have now been fixed: the momentum threshold is applied to the softest of the  $N$  pions (so all  $N$  are above threshold) and the opening angle to the leading pion. The correction removed the spurious ordering-dependent  $0.175 \text{ GeV}/c$  cut on an arbitrary pion, so the well-defined signal count rises (e.g.  $3251 \rightarrow 3746$  for  $2\pi$ ) and the efficiency settles at  $15.5\%$  ( $2\pi$ ) /  $6.1\%$  ( $3\pi$ ) in the corrected phase space (Table 2), slightly below the scan projection because the fix broadens the signal to include soft-leading and wide-angle events that the arbitrary cut had randomly rejected. We verified that the single-pion selection is bit-for-bit unchanged by the fix (identical signal, selected and efficiency on the Run-1 FHC overlay).

## 8 Backgrounds

At these multiplicities the beam-off cosmic (EXT) contribution is expected to dominate the background, followed by  $\nu_{\mu}$  CC events of the wrong charged-pion multiplicity (single-pion events with an extra mis-identified track, or higher-multiplicity events losing a pion) and neutral-current interactions with charged-pion-like tracks. The cut-flow of Table 2 is the  $\nu_{\mu}$  overlay only and therefore does not yet include EXT or dirt; a full background composition, including the data-driven EXT subtraction inherited from the parent analysis, is deferred to the next revision. The MC background composition and any dedicated multi-pion background-control sidebands will be reported alongside.

## 9 Feasibility and observables

With of order 580 selected signal events at full exposure, the two-pion channel can support a flux-averaged total cross section and, plausibly, a coarse (two- to three-bin) differential measurement in a robust observable such as the leading-pion momentum or the total visible hadronic energy. The three-pion channel, with of order 40 selected signal events, is a flux-averaged total-cross-section measurement. Both will use the same Wiener-SVD unfolding and systematic treatment as the parent analysis; the reduced statistics argue for coarse binning and, for  $3\pi$ , a single integrated bin. A concrete observable set and binning will be fixed once the threshold study defines the signal phase space and the background model is in place.

## 10 Systematic uncertainties

The systematic framework — flux (PPFX), cross-section (GENIE reweight), reinteraction and detector-variation universes, propagated through the response matrix and unfolding — is inherited unchanged from the single-pion note. The multi-pion channels are expected to be statistics-limited, so a full systematic evaluation is deferred until the selection and phase space are finalised; the detector-variation samples and the flux/cross-section universes are already available in the same processed files.

## 11 Status and follow-up

- Selections CC1mu2pi / CC1mu3pi implemented, compiled and validated on Run-1 FHC; cut-flow, efficiency and purity established (Section 5).
- Pion-momentum threshold study in progress (Section 6); will fix the accessible phase space and the per-pion threshold, and correct the signal-def bookkeeping for  $N > 1$ .
- To do: full background composition including EXT and dirt; extension to full FHC, RHC and combined exposures; observable/binning choice; total (and, for  $2\pi$ , coarse differential) cross-section extraction with the inherited unfolding and systematics.

## References

- [1] T. Mahmud, *Study of  $\nu_\mu$  charged-current three-charged-pion production in MicroBooNE* (CC3 $\pi^+$  selection), MSc by Research thesis, Lancaster University.
- [2] L. Gu, multi-pion event-selection analyzer for the MicroBooNE NuMI cross-section framework (private communication / internal code).
- [3] I. Pophale and J. Nowak, *Measurement of  $\nu_\mu + \bar{\nu}_\mu$  charged-current single-pion production differential cross sections on argon with MicroBooNE in the NuMI beam*, MicroBooNE analysis note (companion document).